

Fundamentals of Engineering Economics and Management

ENEL301 S2 2026

15 te hua | 15 points

Course co-ordinator:

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Whakamahuki | Course Description:

Technical engineering skills are the foundation of engineering solutions, but they are no use if they are not applied in efficient and effective ways. Successful engineering projects and ventures require management. This course identifies the different management activities involved and develops the skills necessary for managing technology projects. Professional engineers also need to be able to enhance their technical knowledge with essential workplace skills and a wider understanding of current societal issues if they are to be successful in the global workplace. This course aims to improve your understanding of what it means to be a professional engineer. It also aims to give you an introduction to the skills to evaluate technical solutions from ethical, sustainable and economic points of view.

Engineering is a socio-technical enterprise. The technical aspect of engineering focuses on the application scientific and mathematical principle, whilst the social aspect involves ensuring that engineers have a grounded understanding of people and society. Engineering projects and ventures also require management of resources (including time and money) whilst accounting for risk and uncertainty. In summary, engineers reliably conceive, develop, deliver and sustain the technologies that meet human needs, and in doing so, create confidence for people to invest sufficient resources for these endeavours (Trevelyan and Williams, 2018).

This course has been designed for you to gain a broader range of skills and knowledge for you to create value in your engineering solutions, and for you to develop an understanding of the professional responsibilities and duties expected of you as an engineer.

Hua Ako | Course Learning Outcomes:

At the conclusion of this course, you should be able to:

- LO1: Apply ethical principles within engineering contexts, and recognise the social and cultural factors which affect ethics. (WA6, WA8)
- LO2: Apply individual management and inclusive teamwork practices in New-Zealand and global engineering contexts. (WA9, WA10)
- LO3: Conduct financial analysis in engineering contexts, including estimation of cost, cash-flow, financial balances, and financial viability. (WA11)
- LO4: Evaluate the viability of engineering projects, considering social, ethical, national and international laws, environmental, and cultural factors. (WA6, WA7, WA8)
- LO5: Implement engineering project and risk management techniques, including identification of- and communication with-, diverse stakeholders (WA10, WA11)

Mahi ā-Ākonga | Workload

(expected distribution of student hours, note 15 points = 150 hours):

Activity	Time (hours)
Contact Hours	
Lectures	24
Tutorials	0
Workshops	12
Laboratories	0
Independent study	
Review of lectures	24
Test and exam preparation	36
Assignments	54
Tutorial preparation	0
Total	150

Aromatawai | Assessment (method, weight, date due)

Assessment	% of final grade	Date	Course Learning Outcomes	UC Graduate Profile	Washington Accord
Bicultural Workshop	1	Wednesday 22 July 15:00 – 17:00 (B01 – 07) OR Wednesday 29 July 15:00 – 17:00 (B08 – 14)	LO1	GA, BCC	WA6, WA10
Pepeha	4	Friday 31 July 17:00 (B01 – 07) OR Friday 07 August 17:00 (B08 – 14)	LO1-LO2	GA, BCC	WA10
Test	25	Tuesday 08 September 18:30 – 19:30	LO1, LO4	GA, EIE	WA6, WA8
Sustainability assignment (group assessment)	20	Friday 09 October, 17:00	LO4	GA, EIE	WA6, WA7, WA9
Exam	50	TBA	LO1, LO3- LO5	GA, EIE	WA6, WA7, WA8, WA11
Total	100				

The examiners will award a failing grade to students who score less than 40% for the Tests and Exam combined. More formally, $(\text{TestPercent} * 0.25 + \text{ExamPercent} * 0.5) / 0.75$ must be greater than or equal to 40 for a pass mark to be awarded. This note is put in place to ensure that each student has adequately shown to the examiners they have gained some mastery of the topic.

Bicultural Workshop (1%)

The understanding of the need for bicultural competence, and the ability to present oneself with a formal introduction in te reo Māori are becoming essential skills when working as a professional engineering in Aotearoa New Zealand. To support this need, you are required to attend their timetabled Bicultural Workshop. Non-attendance will result in a mark of zero.

Pepeha (4%)

The assignment (4%) involves recording a self-introduction (pepeha) in te reo Māori.

Test (25%)

On Tuesday 8 September (Week 7) a test worth 25% of the final grade will take place. The Test is closed book. UC stickered calculators are permitted. The test will cover all content from Weeks 1 – 6.

Sustainability Assignment (20%)

In the assignment, you will work in a randomly allocated team to investigate and report on the sustainability impacts of a product or service system. The investigation will include assessment of the potential environmental, social, and economic outcomes across the life cycle.

Exam (50%)

In the exam period, the Exam worth 50% of the final grade will take place. The Exam is closed book. UC stickered calculators are permitted. The exam will cover all content of the course.

Lateness Penalties

The Bicultural Workshop attendance is compulsory; if you do not attend you will be marked zero. For the Pepeha and Sustainability Assignment, a lateness penalty of 10% (in absolute terms) per day or part day late will be deducted from the original mark. For example, an assignment with a nominal mark of 83% submitted 0-24 hours late will receive a mark of 73%, and submitted 24-48 hours late will receive 63%.

Course Breakdown

Term 3			
Week beginning	Lecture material	Assessment	Workshops
13 July	Introduction to Engineering Practice & Ethics Collaboration & Teamwork		Ethics
20 July	Māori Perspectives in Engineering Bias, Diversity & Ethics	BiCC workshop (1%) – Tutorials B01-07	BiCC workshop (timetable and attend one of B01-07 OR B08-14)
27 July	Intellectual property	BiCC workshop (1%) – Tutorials B08-14 Pepeha (4%) – Tutorials B01-07	
3 August	Law	Pepeha (4%) – Tutorials B08-14	IP Protection
10 August	Data Privacy and Data Ethics		Law, Data Privacy, Data Ethics

	Business Structures, Strategy, Decisions & Risk		
17 August	Sustainable Development & Systems Thinking Life Cycle Assessment (1/2)		Sustainable Development & Systems Thinking

Term 4			
Week beginning	Lecture material	Assessment	Workshops
7 September	Life Cycle Assessment (2/2) Social and Economic Assessment	Test (25%). Refer to your timetable for room allocation	Life Cycle Assessment
14 September	Time Value of Money & Cash Flow Cost Accounting		Time Value of Money & Cash Flow
21 September	Financial Accounting & Reporting		Cost Accounting, Financial Accounting, Financial Reporting
28 September	Project Management Risk Management		Project Management
5 October	Opportunity Analysis Guest Lecturer	Sustainability Assignment (20%)	Support for Sustainability Assignment
13 October	Integrated Decisions Course Summary		Risk Management

Tuhinga | Texts and Readings (recommendations):

Trevelyan, J. (2021). Learning Engineering Practice (1st ed.). CRC Press. <https://doi.org/10.1201/b22622>.
[Hyperlink to library \(if this fails, search manually through UC library\)](#)

Trevelyan, J. (2014). The Making of an Expert Engineer (1st ed.). CRC Press. <https://doi.org/10.1201/b17434>.
[Hyperlink to library \(if this fails, search manually through UC library\)](#)

Herenga Akoranga | Academic Policies (e.g. special consideration, dishonest practice):

Scaling of marks

In order to maintain consistency across courses and fairness for students, scaling of raw marks occurs. In the Faculty of Engineering, target course GPAs are calculated based on the performance of the cohort of students in their courses in the previous year. Scaling of the raw total course marks is normally performed so that when converted to grades (using UC Grade Scale) the outgoing GPA is in line with the target GPA for a course. Scaling up or down can occur.

The Grading Scale for the University:

<https://www.canterbury.ac.nz/study/study-support-info/study-related-topics/grading-scale>

Artificial Intelligence Tools

The use of Artificial Intelligence (AI) tools for each of the assessments in ENEL301 is summarised in the Table below. No AI use is allowed in the tests and exam because these are closed-book invigilated assessments. Students are always responsible for the accuracy of the submitted works, regardless of which tools are used.

Assessment Item	Permitted use of AI	
Bicultural Workshop	Not applicable – attendance only	
Pepeha	Generative AI tools are not restricted for this assessment.	
Sustainability Assignment	Generative AI Tools Are Permitted for Certain Parts of This Assessment	
Tests	Generative AI tools cannot be used for this assessment.	
Exam	Generative AI tools cannot be used for this assessment.	

Generative AI Tools Are Permitted for Certain Parts of This Assessment

In these assessments (Sustainability Assignment), you are permitted to use generative artificial intelligence (AI) for the purpose of proof reading and editing the document, and for gathering and summarising knowledge. No other use of generative AI is permitted. To assist with maintaining academic integrity, you must appropriately acknowledge any use of generative AI in your work. Please include a Statement of AI use (if no AI tool has been used, then this must also be stated) and a listing of all prompts provided to the AI tool, clearly indicating which AI tools were used and how they contributed to your assessment.

Special Considerations

As per the Faculty of Engineering policy, the Professional years of the BE(Hons) degree no longer use computed grades as a remedy for special considerations.

For the invigilated assessments (the Test and Exam), the default remedy will be a delayed or repeated sitting of an alternative assessment item that shows equivalent learning outcomes; if a delayed or repeated sitting is not appropriate and evidence shows that the course learning outcomes are already met, the student may be awarded a pass grade (P AEG).

In the case of a re-sit of either the Test or the Exam, the grade in the second attempt only is used to calculate the final grade. The Exam resit will take place in the Week of Monday November 23rd to Friday November 27th. Students should inform the Course Coordinator if they have applied for Special Consideration for the Tests or Exam.

For the Pepeha and Assignment, the default remedy will be an extension to the submission deadline. Students should apply directly to the Course Co-ordinator if they meet the Special Considerations criteria (with supporting evidence e.g. medical certificate) rather than through the Special Consideration process. Extensions cannot be granted for having several assessments due at the same time, mistaking the due date or time, or other work commitments. For further information on Special Considerations, refer to: <https://www.canterbury.ac.nz/study/special-consideration/>.

Assessment Completion

Students must attempt all assessment items in order to be eligible to pass the course as defined in the General Conditions for Credit Regulations 3(a) [here](#).

Assessment and Grade Appeals

Within five working days of receiving the results of, or feedback for, an assessment (whichever is later), a student may apply to the Course Coordinator to have the assessment reconsidered. [Taught and Project Course Assessment Regulation 7(a) [here](#)]

Within four weeks of the final course grades being released, a student may apply for reconsideration of a final grade on the appropriate form. Only assessment items that have not already been returned to students will be fully re-marked, while the marks of previous assessments will be checked. Note that a fee is charged, but this is refunded if a change of grade occurs. [Taught and Project Course Assessment Regulation 7(d) [here](#)]

For further information on a reconsideration of grade, see here <https://www.canterbury.ac.nz/study/study-support-info/study-related-topics/examinations/exam-results-dates-and-appeals>

Emergency Provisions

In the case of an emergency that affects the whole course, the Course Coordinator, in consultation with the Dean, may change the nature, weighting and timing of assessments, e.g. tests and examination may be replaced with assignments of the same weight or different weight at a different time and/or date (which, under certain circumstances, may be outside the prescribed course dates). The 'Special consideration' process will also be used for unforeseen circumstances that adversely affect the academic performance of students individually. The usual grounds for this are described in the UC policy 'Special Consideration Procedures and Guidelines', and personal circumstances due to a wider emergency event may also qualify

Dishonest Practice

Plagiarism, collusion, copying and ghost writing are unacceptable and dishonest practices.

- Plagiarism is the presentation of any material (test, data, figures or drawings, on any medium including computer files) from any other source without clear and adequate acknowledgment of the source.
- Collusion is the presentation of work performed in conjunction with another person or persons, but submitted as if it has been completed only by the named author(s).
- Copying is the use of material (in any medium, including computer files) produced by another person(s) with or without their knowledge and approval.
- Ghost writing is the use of another person(s) (with or without payment) to prepare all or part of an item submitted for assessment.

Do not engage in dishonest practices. The Department reserves the right to refer dishonest practices to the University Proctor and where appropriate to not mark the work.

Harassment of any sort will not be tolerated

Each student is here at UC to learn and to experience a friendly and supportive community. It is every student's right to expect: respect and courtesy from staff and other students, including freedom from harassment of any sort; fair treatment; the ability to speak out about any issues that concern them, without fear of consequences for their safety and well-being. Furthermore, each student has the responsibility to: respect the rights and property of others; attend to their own health and safety and that of others; and behave in a manner towards each other that does not reflect badly on the student body or the University.

For more information, or to find out where help is available, refer to:
www.canterbury.ac.nz/ucpolicy/GetPolicy.aspx?file=Harassment-Policy.pdf.

UC Graduate Profile

EIE	Employable, Innovative & Enterprising • Working effectively and professionally with diverse communities • Communication • Analytical, critical thinking and problem solving in diverse contexts • Digital literacy • Innovation, enterprising and creativity	x
BCC	Bicultural Competence & Confidence • A process of self-reflection on the nature of 'knowledge' and 'norms' • The nature of contemporary Māori organisational structures, e.g. rūnanga, hapū, iwi, iwi corporations • Traditional and contemporary realities of Māori society, e.g. tikanga and kawa, te reo Māori • The Treaty of Waitangi and Aotearoa New Zealand's bicultural history • The processes of colonisation and globalisation • Other Indigenous models of development, knowledge and behaviours • Application of bicultural competence and confidence in a chosen discipline and career	x
GA	Globally Aware • Self-reflection on the nature of one's culture, language and beliefs and on one's systems of knowledge • Understanding the global nature of one's discipline • The ability to engage critically and effectively in global and multicultural contexts	
CE	Community Engagement • Engagement: Gaining knowledge and understanding of a community by interacting with a community • Reflection: Gaining knowledge and understanding of a community through reflection on one's experience with that community • Application: Understanding and articulating how the content and/or skills of the subject/programme enhances the community	

Washington Accord

Blank (none)

*Underpinning knowledge is provided

**Basic learning occurs and is assessed

***More advanced learning occurs

****More advanced learning occurs and is robustly assessed

Washington Accord Outcomes		Level
WA1	Engineering knowledge	**
	Apply knowledge of mathematics, natural science, computing and engineering fundamentals, and an engineering specialization as specified in WK1 to WK4 respectively to develop solutions to complex engineering problems.	
WA2	Problem analysis	**
	Identify, formulate, research literature and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences and engineering sciences with holistic considerations for sustainable development* (WK1 to WK4). *	
WA3	Design / development of solutions	*
	WA3: Design creative solutions for complex engineering problems and design systems, components or processes to meet identified needs with appropriate consideration for public health and safety, whole-life cost, net zero carbon as well as resource, cultural, societal, and environmental considerations as required (WK5)	
WA4	Investigation	**
	Conduct investigations of complex engineering problems using research methods including research based knowledge, design of experiments, analysis and interpretation of data, and synthesis of information to provide valid conclusions (WK8)	
WA5	Tool usage	
	Create, select and apply, and recognize limitations of appropriate techniques, resources, and modern engineering and IT tools, including prediction and modelling, to complex engineering problems (WK2 and WK6)	
WA6	The Engineer and The World	***
	When solving complex engineering problems, analyze and evaluate sustainable development impact (Represented by the 17 UN Sustainable Development Goals (UN-SDG) to: society, the economy, sustainability, health and safety, legal frameworks, and the environment (WK1, WK5, and WK7)	
WA7	Ethics	****
	Apply ethical principles and commit to professional ethics and norms of engineering practice and adhere to relevant national and international laws. Demonstrate an understanding of the need for diversity and inclusion (WK9)	
WA8	Individual and Collaborative Team work	***
	Function effectively as an individual, and as a member or leader in diverse and inclusive teams and in multi-disciplinary, face-to-face, remote and distributed settings (WK9)	

Blank (none)

*Underpinning knowledge is provided

**Basic learning occurs and is assessed

***More advanced learning occurs

****More advanced learning occurs and is robustly assessed

Washington Accord Outcomes		Level
WA9	Communication	**
	Communicate effectively and inclusively on complex engineering activities with the engineering community and with society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations, taking into account cultural, language, and learning differences	
WA10	Project management and finance	****
	Apply knowledge and understanding of engineering management principles and economic decision-making and apply these to one's own work, as a member and leader in a team, and to manage projects and in multidisciplinary environments.	
WA11	Lifelong learning	*
	Recognize the need for, and have the preparation and ability for i) independent and life-long learning ii) adaptability to new and emerging technologies and iii) critical thinking in the broadest context of technological change (WK8)	

Washington Accord Knowledge Profile

WK1: A systematic, theory-based understanding of the natural sciences applicable to the discipline and awareness of relevant social sciences.

WK2: Conceptually-based mathematics, numerical analysis, data analysis, statistics and formal aspects of computer and information science to support detailed analysis and modelling applicable to the discipline.

WK3: A systematic, theory-based formulation of engineering fundamentals required in the engineering discipline.

WK4: Engineering specialist knowledge that provides theoretical frameworks and bodies of knowledge for the accepted practice areas in the engineering discipline; much is at the forefront of the discipline.

WK5: Knowledge, including efficient resource use, environmental impacts, whole-life cost, re-use of resources, net zero carbon, and similar concepts, that supports engineering design and operations in a practice area.

WK6: Knowledge of engineering practice (technology) in the practice areas in the engineering discipline.

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WK7: Knowledge of the role of engineering in society and identified issues in engineering practice in the discipline, such as the professional responsibility of an engineer to public safety and sustainable development.

WK8: Engagement with selected knowledge in the current research literature of the discipline, awareness of the power of critical thinking and creative approaches to evaluate emerging issues.

WK9: Ethics, inclusive behaviour and conduct. Knowledge of professional ethics, responsibilities, and norms of engineering practice. Awareness of the need for diversity by reason of ethnicity, gender, age, physical ability etc. with mutual understanding and respect, and of inclusive attitudes.